WORKED SOLUTION

STEP I 2004, Question 2

Pure • Floor Function Integrals

STEP I 2004, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q2

The key idea

The function $[x]$ is constant on each interval $r \leq x < r+1$, so anything built from it is a step function. Integrating a step function is just adding up areas of rectangles: split $[0, a]$ at the integers, and on each unit piece the integrand is a constant. When a is not an integer, the only new feature is a final, narrower rectangle.

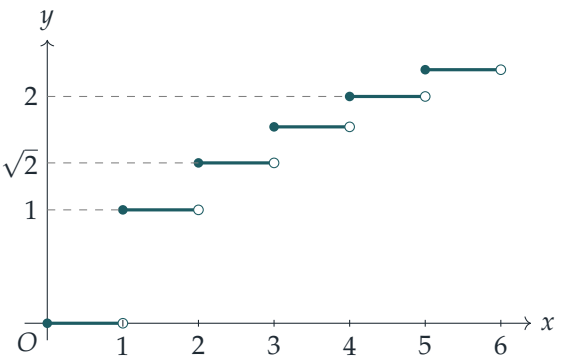
Solution

Part (i)

For $x \geq 0$, write $r = [x]$. Then r is a non-negative integer and $r \leq x < r+1$, so on that interval

$$y = \sqrt{[x]} = \sqrt{r}.$$

So the graph is a staircase: $y = 0$ on $[0,1)$, $y = 1$ on $[1,2)$, $y = \sqrt{2}$ on $[2,3)$, and so on. Each step includes its left-hand endpoint (solid dot) but not its right-hand one (open dot), since $[x]$ jumps up exactly at each integer. The steps rise by ever smaller amounts, as $\sqrt{r+1} - \sqrt{r} = 1/(\sqrt{r+1} + \sqrt{r})$ decreases. For $x < 0$ we have $[x] \leq -1$, so $\sqrt{[x]}$ is not defined there and the graph is only for $x \geq 0$.



Now let a be a positive integer. Split the range of integration at the integers $1, 2, \ldots, a-1$:

$$\int_0^a \sqrt{[x]}\,\mathrm{d}x = \sum_{r=0}^{a-1} \int_r^{r+1} \sqrt{[x]}\,\mathrm{d}x.$$

On $r \le x < r+1$ the integrand equals the constant $\sqrt{r}$. (At the single point $x = r+1$ it takes a different value, but changing a function at one point does not change its integral.) So each piece is the area of a rectangle of width 1 and height $\sqrt{r}$:

$$\int_r^{r+1} \sqrt{[x]}\,\mathrm{d}x = \sqrt{r}, \qquad \text{and hence} \qquad \int_0^a \sqrt{[x]}\,\mathrm{d}x = \sum_{r=0}^{a-1} \sqrt{r},$$

as required.

Part (ii)

Exactly the same argument works for $2^{[x]}$, which equals 2^r on $r \le x < r+1$. For a positive integer a,

$$\int_0^a 2^{[x]}\,\mathrm{d}x = \sum_{r=0}^{a-1} \int_r^{r+1} 2^r\,\mathrm{d}x = \sum_{r=0}^{a-1} 2^r = 1 + 2 + 4 + \cdots + 2^{a-1}.$$

This is a geometric series with first term 1, common ratio 2 and a terms, so its sum is

$$\frac{2^a - 1}{2 - 1} = 2^a - 1,$$

which is the required result.

Part (iii)

Now let $a > 0$ not be an integer, and put $n = [a]$, so n is an integer with $n < a < n+1$ and $n \ge 0$.

Ask yourself Where does the integral stop being a sum of whole unit rectangles?

Up to $x = n$ everything is as in part (ii); the leftover piece is $n \le x \le a$, a strip of width $a - n$ lying inside a single step, where $[x] = n$. (If $n = 0$ the first piece is empty and contributes $2^0 - 1 = 0$, so the same formula works.) Therefore

$$\begin{aligned}\int_0^a 2^{[x]}\,\mathrm{d}x &= \int_0^n 2^{[x]}\,\mathrm{d}x + \int_n^a 2^n\,\mathrm{d}x \\ &= (2^n - 1) + (a - n)\,2^n \\ &= 2^{[a]}(1 + a - [a]) - 1.\end{aligned}$$

Checks

- If $0 < a < 1$, then $[a] = 0$ and the formula gives $1 \cdot (1 + a) - 1 = a$: correct, since the integrand is 1 on $[0, a]$.

- As a approaches an integer m from below, $[a] = m - 1$ and the formula tends to $2^{m-1} \cdot 2 - 1 = 2^m - 1$, matching part (ii). So the integral is a continuous function of a, as any integral of a bounded function with respect to its upper limit must be.
- Numerically, with $a = 2.5$ the formula gives $4 \times 1.5 - 1 = 5$, and a midpoint-rule evaluation of the integral also gives 5.000. For part (i) with $a = 5$, both sides are $0 + 1 + \sqrt{2} + \sqrt{3} + 2 \approx 6.1463$.

Common slips

- Summing from $r = 1$ to a instead of $r = 0$ to $a - 1$: on $[r, r+1)$ the height is $\sqrt{r}$, the value at the *left* end.
- In part (iii), forgetting the partial strip, or giving it width 1 rather than $a - [a]$.
- Drawing the steps joined by vertical lines, or with closed dots at both ends: the floor function is not continuous at the integers.

Beyond STEP

Integrating a step function by summing rectangles is exactly how the Riemann integral is built at university: one defines the integral for step functions first, then squeezes a general function between step functions from above and below. Part (i) is also the first move in comparing sums with integrals, which gives estimates such as $\sum_{r=1}^{n} \sqrt{r} \approx \frac{2}{3} n^{3/2}$. Abbott's *Understanding Analysis* develops the Riemann integral in this way.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.